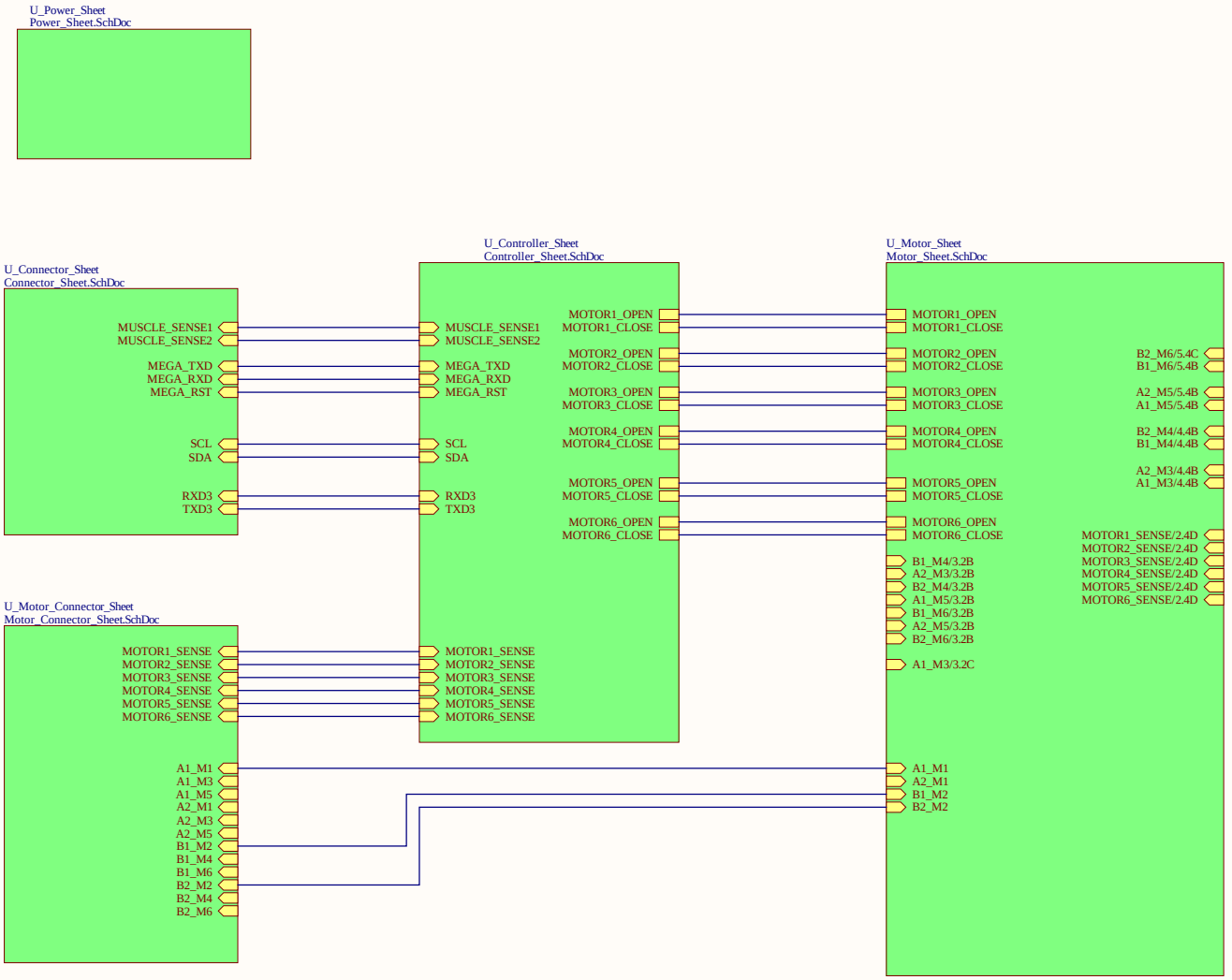
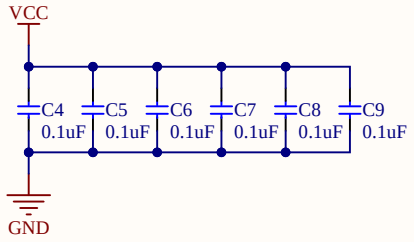
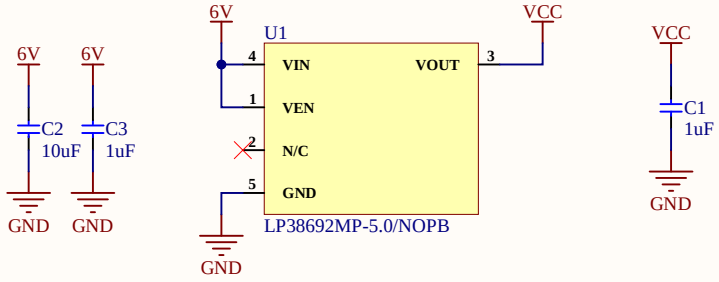

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
MEGA_TXD
MEGA_RXD
MEGA_RST
SCL
SDA
RXD3
TXD3
U_Controller_Sheet
Controller_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
MEGA_TXD
MEGA_RXD
MEGA_RST
SCL
SDA
RXD3
TXD3
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
U_Motor_Sheet
Motor_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
B1_M4/3.2B
A2_M3/3.2B
B2_M4/3.2B
A1_M5/3.2B
B1_M6/3.2B
A2_M5/3.2B
B2_M6/3.2B
A1_M3/3.2C
B2_M6/5.4C
B1_M6/5.4B
A2_M5/5.4B
A1_M5/5.4B
B2_M4/4.4B
B1_M4/4.4B
A2_M3/4.4B
A1_M3/4.4B
MOTOR1_SENSE/2.4D
MOTOR2_SENSE/2.4D
MOTOR3_SENSE/2.4D
MOTOR4_SENSE/2.4D
MOTOR5_SENSE/2.4D
MOTOR6_SENSE/2.4D
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
A1_M1
A2_M1
B1_M2
B2_M2

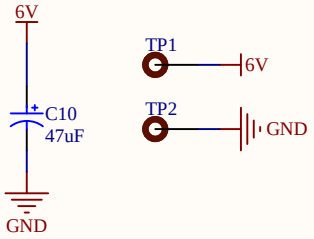
DECOUPLING CAPACITORS



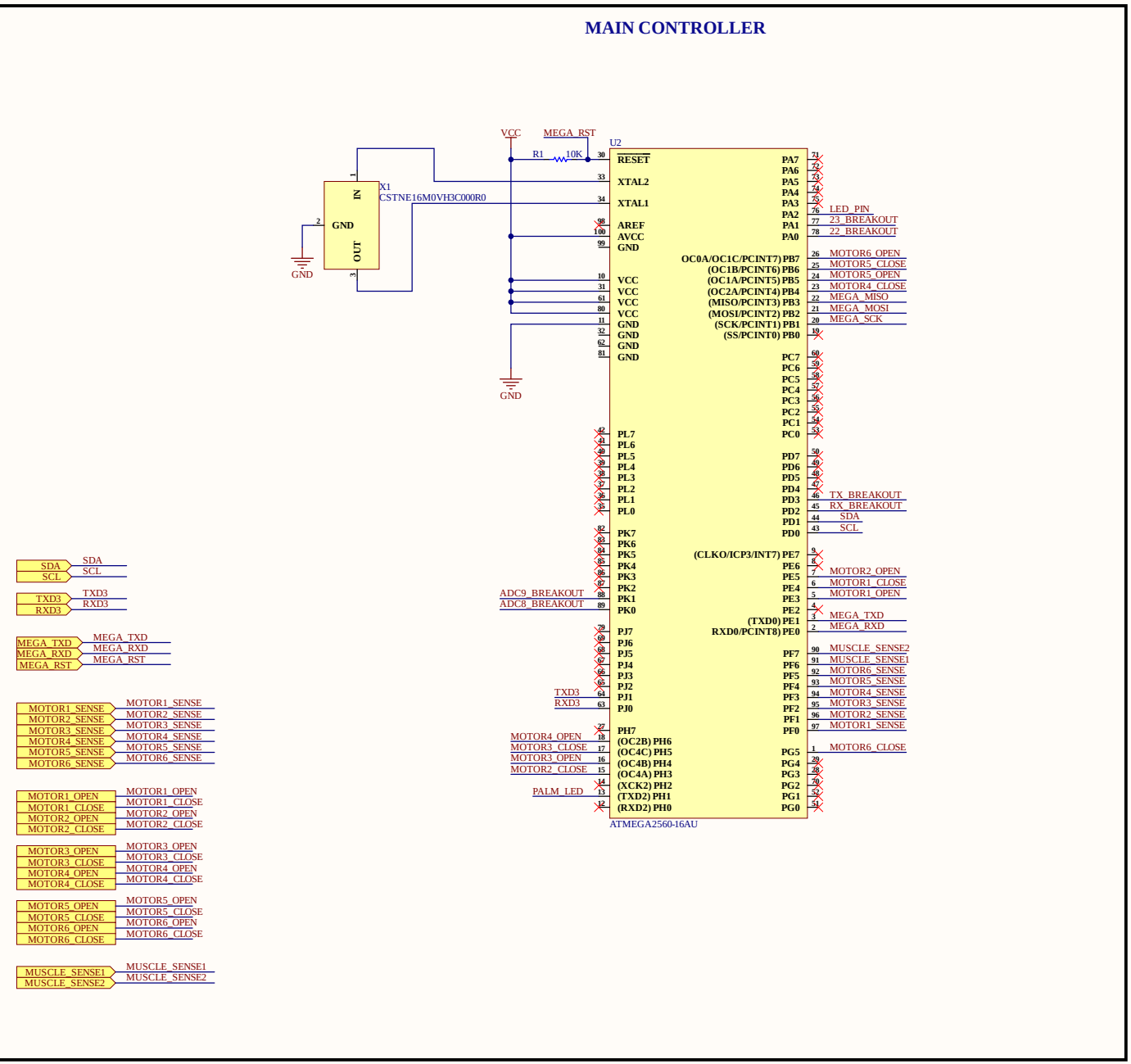
6V TO 5V CONVETER



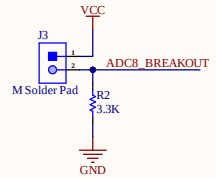
POWER INPUTS



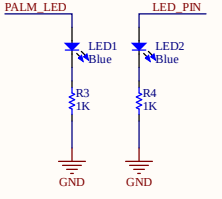
MAIN CONTROLLER



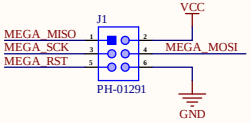
SENSOR CONNECTION

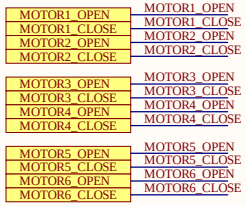


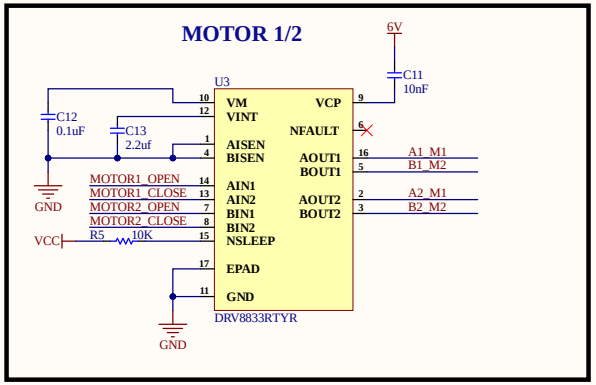
LED'S

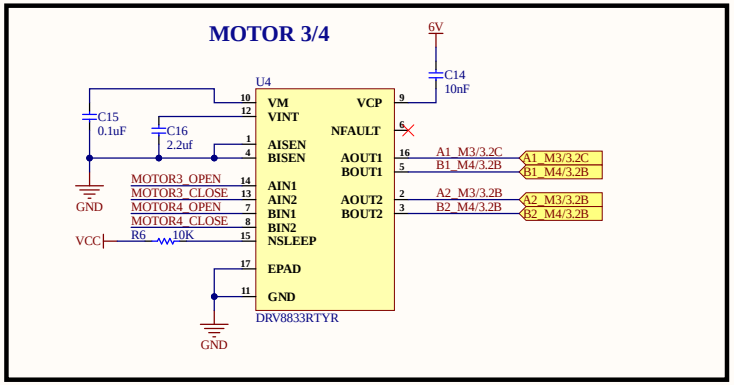


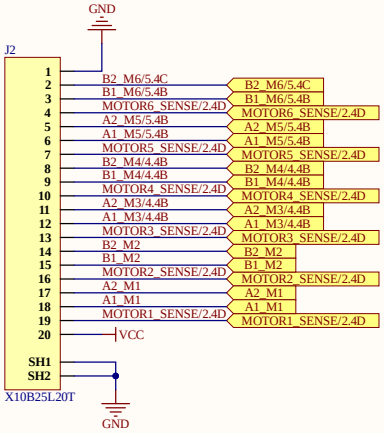
SPI

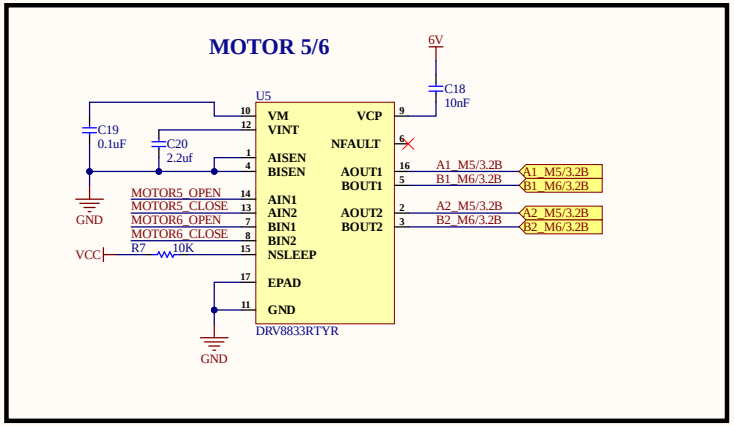


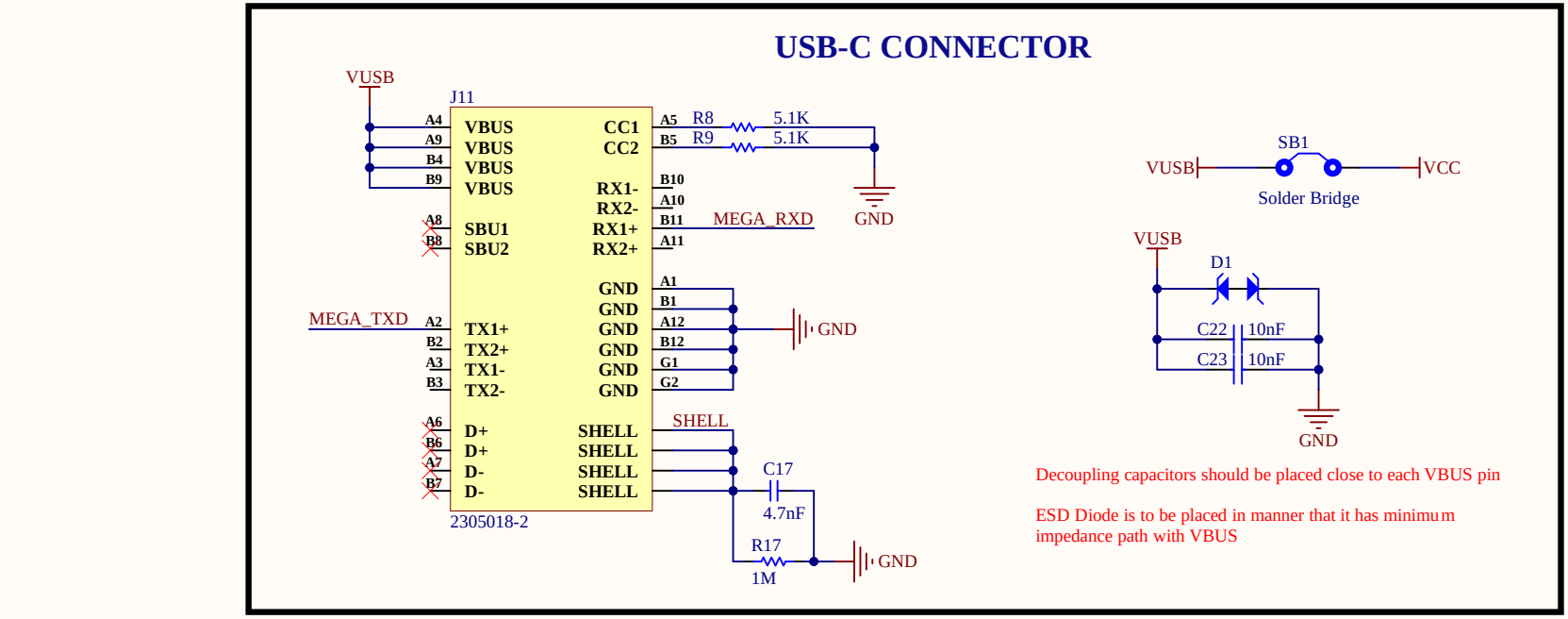

MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE

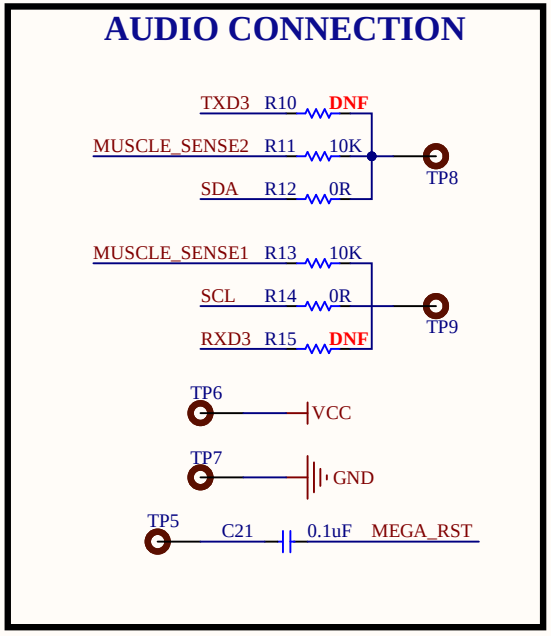

MOTOR 1/2
6V
U3
C11
10nF
C12
0.1uF
C13
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
AIN1
AIN2
BIN1
BIN2
NSLEEP
AOUT2
BOUT2
EPAD
GND
DRV8833RTYR
A1_M1
B1_M2
A2_M1
B2_M2
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
R5
10K
VCC
GND

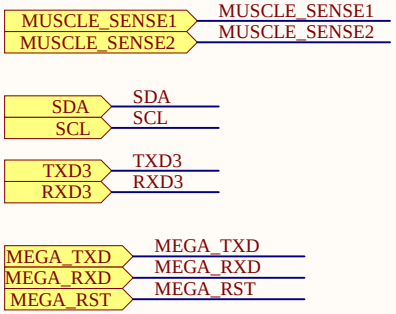

MOTOR 3/4
6V
U4
C14
10nF
C15
0.1uF
C16
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
AIN1
AIN2
BIN1
BIN2
NSLEEP
AOUT2
BOUT2
EPAD
GND
DRV8833RTYR
A1_M3/3.2C
B1_M4/3.2B
A2_M3/3.2B
B2_M4/3.2B
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
R6
10K
VCC
GND

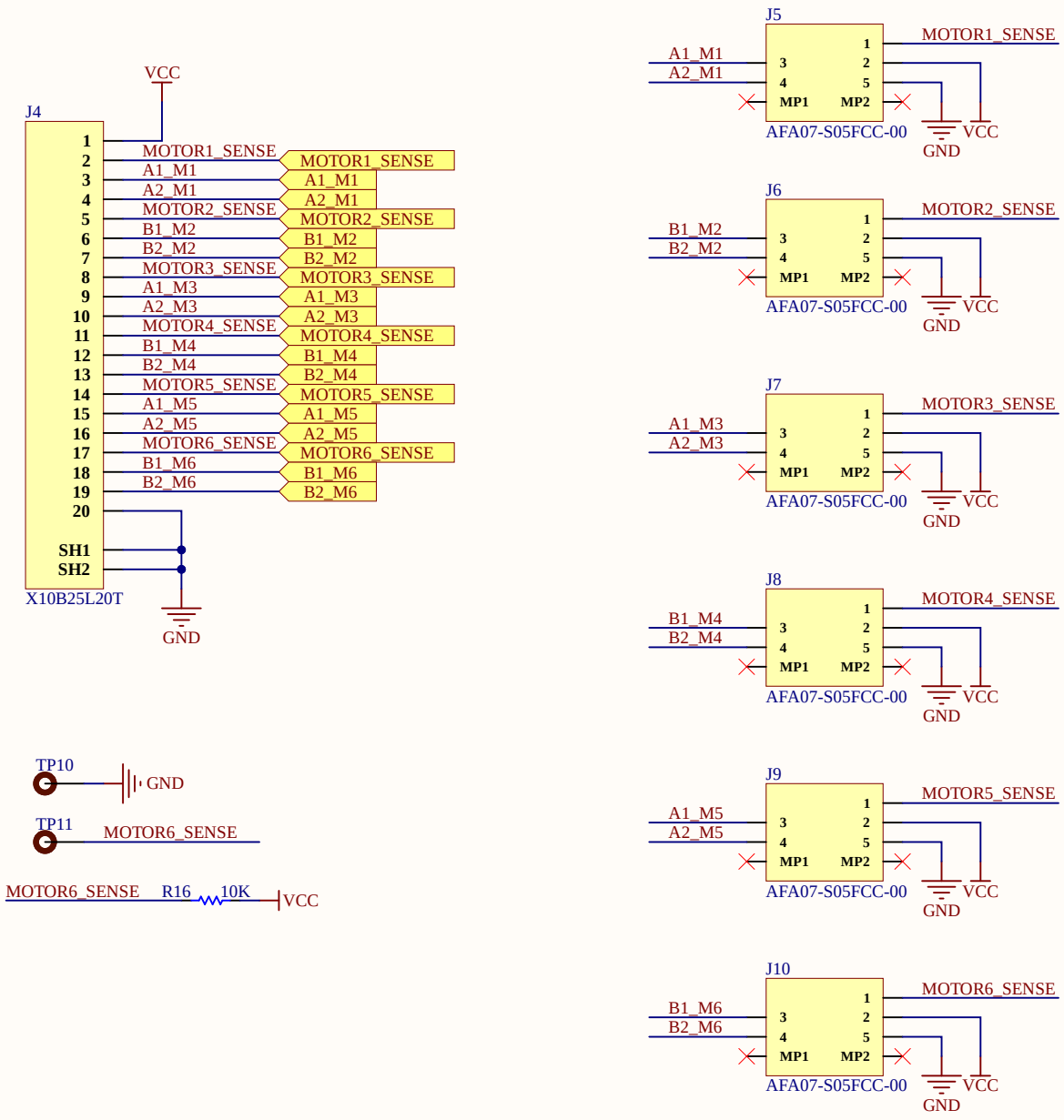

GND
J2
B2_M6/5.4C
B1_M6/5.4B
MOTOR6_SENSE/2.4D
A2_M5/5.4B
A1_M5/5.4B
MOTOR5_SENSE/2.4D
B2_M4/4.4B
B1_M4/4.4B
MOTOR4_SENSE/2.4D
A2_M3/4.4B
A1_M3/4.4B
MOTOR3_SENSE/2.4D
B2_M2
B1_M2
MOTOR2_SENSE/2.4D
A2_M1
A1_M1
MOTOR1_SENSE/2.4D
VCC
SH1
SH2
X10B25L20T
GND


MOTOR 5/6
6V
U5
C18
10nF
C19
0.1uF
C20
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
AIN1
AIN2
BIN1
BIN2
NSLEEP
AOUT2
BOUT2
EPAD
GND
DRV8833RTYR
A1_M5/3.2B
B1_M6/3.2B
A2_M5/3.2B
B2_M6/3.2B
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
R7
10K
VCC
GND


USB-C CONNECTOR
VUSB
J11
A4
B9
B4
A9
VBUS
VBUS
VBUS
VBUS
CC1
CC2
RX1-
RX2-
RX1+
RX2+
A5
B5
B10
A10
B11
A11
R8
R9
5.1K
5.1K
GND
MEGA_RXD
A8
B8
SBU1
SBU2
MEGA_TXD
A2
B2
A3
B3
TX1+
TX2+
TX1-
TX2-
GND
GND
GND
GND
A1
B1
A12
B12
G1
G2
GND
A6
B6
A7
B7
D+
D+
D-
D-
SHELL
SHELL
SHELL
SHELL
SHELL
C17
4.7nF
R17
1M
GND
2305018-2
SB1
VUSB
VCC
Solder Bridge
VUSB
D1
C22
10nF
C23
10nF
GND
Decoupling capacitors should be placed close to each VBUS pin
ESD Diode is to be placed in manner that it has minimum impedance path with VBUS


AUDIO CONNECTION
TXD3
R10
DNF
MUSCLE_SENSE2
R11
10K
TP8
SDA
R12
0R
MUSCLE_SENSE1
R13
10K
SCL
R14
0R
TP9
RXD3
R15
DNF
TP6
VCC
TP7
GND
TP5
C21
0.1uF
MEGA_RST


MUSCLE_SENSE1
MUSCLE_SENSE2
MUSCLE_SENSE1
MUSCLE_SENSE2
SDA
SCL
SDA
SCL
TXD3
RXD3
TXD3
RXD3
MEGA_TXD
MEGA_RXD
MEGA_RST
MEGA_TXD
MEGA_RXD
MEGA_RST


VCC
J4
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
SH1
SH2
X10B25L20T
MOTOR1_SENSE
A1_M1
A2_M1
MOTOR2_SENSE
B1_M2
B2_M2
MOTOR3_SENSE
A1_M3
A2_M3
MOTOR4_SENSE
B1_M4
B2_M4
MOTOR5_SENSE
A1_M5
A2_M5
MOTOR6_SENSE
B1_M6
B2_M6
GND
TP10
GND
TP11
MOTOR6_SENSE
MOTOR6_SENSE
R16
10K
VCC
J5
A1_M1
A2_M1
MOTOR1_SENSE
MP1
MP2
AFA07-S05FCC-00
VCC
GND
J6
B1_M2
B2_M2
MOTOR2_SENSE
AFA07-S05FCC-00
VCC
GND
J7
A1_M3
A2_M3
MOTOR3_SENSE
AFA07-S05FCC-00
VCC
GND
J8
B1_M4
B2_M4
MOTOR4_SENSE
AFA07-S05FCC-00
VCC
GND
J9
A1_M5
A2_M5
MOTOR5_SENSE
AFA07-S05FCC-00
VCC
GND
J10
B1_M6
B2_M6
MOTOR6_SENSE
AFA07-S05FCC-00
VCC
GND